

Final Year Project Proposal Form

Project Title	Autonomous Driving Agent Training Using Deep Reinforcement Learning in CARLA Simulator
Supervisor Name	Dr. Mohammad Shadab Khan
Co-Supervisor Name (if any)	-
Project Status	Student -Proposed
Industry Collaboration	No
Company Name, Contact Name, Contact Phone (if the answer to Industry Collaboration is Yes)	-
Project Type	Research-Based
Project Specialization (Project Specialisation and Student Specialisation should match)	Data Science
Project Category (Pls. refer at the end of document for the selection of category based on the specialisation)	Data Analytics
Project Focus/Contribution (Pls. refer at the end of document for the selection of focus/contribution)	Analysis of Data

based on the specialisation)	
Project Description (Discuss Background, Problem Statement, Methodology, Expected Output/Significance in summary form)	Background Autonomous driving has become a major trend in the automotive industry, with most leading car manufacturers integrating autonomous capabilities into their vehicles. However, despite rapid progress, autonomous driving systems still face several challenges. Traditional rule-based control methods struggle to handle complex and dynamic real-world environments, such as unpredictable traffic behaviors or varying road conditions. In addition, the training of deep learning-based driving models requires large amounts of labeled data, which are often difficult, expensive, or unsafe to collect in real-world scenarios. To address these issues, simulation environments such as CARLA provide a safe and flexible platform for developing and testing autonomous driving algorithms. By combining simulation with deep reinforcement learning (DRL), it becomes possible to train intelligent driving agents capable of learning optimal driving strategies through interaction with virtual environments. Problem Statement Current autonomous driving systems rely heavily on rule-based control or supervised learning, which often struggle to handle complex and dynamic traffic environments. These methods require large amounts of labeled data and cannot easily adapt to unseen scenarios, leading to poor generalization and unsafe driving behaviors. To address these limitations, this project aims to develop a reinforcement learning-based autonomous driving model trained in the CARLA simulation environment. The approach focuses on improving the decision-making ability and safety of autonomous agents through continuous interaction and learning in a simulated yet realistic environment. Methodology • Literature Review: Conduct a detailed review of research papers on autonomous driving with reinforcement learning to identify limitations and potential improvements.

	• Set up an Autonomous Driving Training Simulation Environment: Set up CARLA simulator to provide realistic road scenarios for model training and testing. • Model Development: Develop models such as CNN and Transformer based on the Proximal Policy Optimization (PPO) algorithm to train the autonomous driving agent. • Reinforcement Learning Training: Train the agent within the simulation environment using PPO to optimize driving behavior such as lane following, obstacle avoidance, and collision reduction. • Performance Evaluation: Evaluate models using key metrics and Carla provided benchmark (No crash benchmark) to identify the most reliable framework. Expected Output / Significance This project aims to develop an intelligent autonomous driving agent capable of stable navigation in CARLA driving scenarios, demonstrating safe obstacle avoidance and effective decision-making.
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Project Objectives (Focused and precise list of statements that can imply the goals to be achieved, Majority of the Project Objectives – using SMART objectives)	1. To set up and configure the CARLA simulation environment for developing and testing autonomous driving models within the first month of the project. 2. To develop and implement a deep reinforcement learning model using the Proximal Policy Optimization (PPO) algorithm to train an autonomous driving agent. 3. To evaluate and compare the performance of different model architectures (e.g., CNN, Transformer) in terms of stability, safety, and navigation efficiency. 4. To analyze the trained agent's driving behavior using key metrics such as success rate, collision rate, and average reward to assess learning effectiveness. 5. To propose an improved reinforcement learning framework for autonomous driving that demonstrates safer and more reliable driving performance in simulated environments.
Project Outcomes (Outcomes are in line with the Project and Student specialisation)	1. A functional autonomous driving model trained using reinforcement learning within the CARLA simulator, capable of stable navigation and obstacle avoidance. 2. A proposed reinforcement learning framework for autonomous driving that demonstrates improved safety and decision-making performance in simulated environments. 3. Comprehensive performance analysis results using key metrics such as No Crash benchmark to validate model effectiveness. 4. A detailed performance evaluation report, including metrics such as success rate, collision rate, and average reward, to assess model effectiveness and reliability. 5. A research report documenting environment setup, model development, training results, performance analysis, and recommendations for future improvements.

Project Scope (Focus/Expected Output/ Deliverables with the limits and constraints of the study can be described and implies enough scope for the two-trimester project)	Focus The project focuses on the development of reinforcement learning models for autonomous driving. It aims to contribute to the field of intelligent transportation through research and experimentation rather than prototype deployment. Expected Output A comprehensive research report and a trained autonomous driving model demonstrating improved driving performance within the CARLA simulator. Limit and Constrains · This study is limited to the CARLA simulation environment and does not involve real-world vehicle testing. · The research focuses solely on software-based modeling, analysis, and experimentation without the integration of any hardware components. · Model accuracy and training performance depend heavily on available computational resources, processing speed, and network stability.
Number of Students (If it is two-students project, subtitles and work distribution must be clearly specified and differentiated for each student)	One
Student 1 Subtitle (Pls. fill up if the number of students is two)	
Student 1 Work Distribution (Pls. fill up if the number of students is two)	

Student 2 Subtitle (Pls. fill up if the number of students is two)	
Student 2 Work Distribution (Pls. fill up if the number of students is two)	
Student 1 Details (Student Name, Student ID, Specialisation, Handphone number, E-mail address) (Leave it blank, if unknown)	1. Name: NG JUN XUAN 2. Student ID: 242UC243LS 3. Specialisation: Data Science 4. Handphone Number: 0103678673 5. Email Address: NG.JUN.XUAN@student.mmu.edu.my

Student 2 Details (if it is a two-student project) (Student Name, Student ID, Specialisation, Handphone Number, E-mail address) (Leave it blank, if unknown)	
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